

DESIGN AND ANALYSIS OF A 230 V AC TO 5 V DC POWER SUPPLY

Detailed Design, Component Selection and Simulation Analysis

Student Name	Sathya T
Course	B.tech in EEE

1. Objective

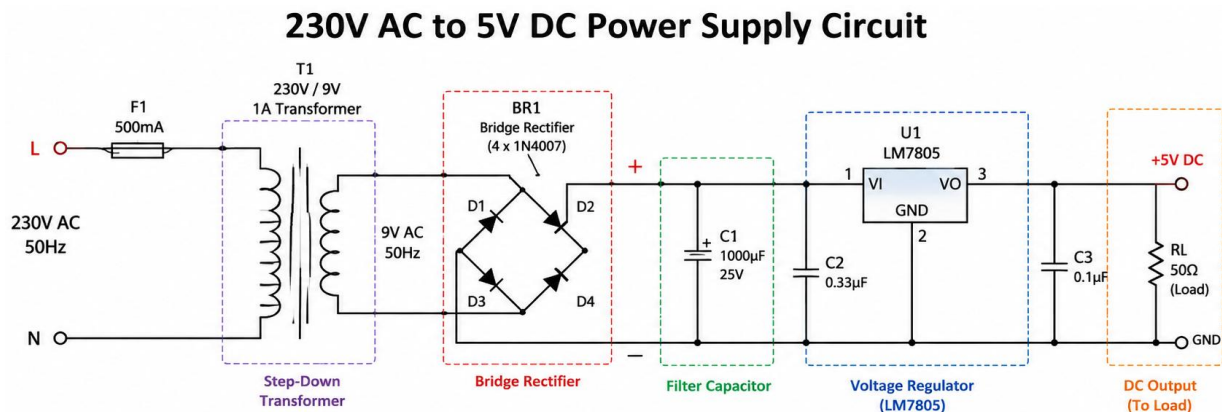
To design and analyze a regulated power supply that converts 230 V AC, 50 Hz mains input into a stable 5 V DC output suitable for low-voltage electronic circuits. The design uses an isolating step-down transformer, full-wave bridge rectifier, smoothing capacitor and LM7805 linear voltage regulator.

2. Design Overview

The proposed supply follows four electrical stages: (1) isolation and voltage reduction, (2) full-wave rectification, (3) capacitive filtering, and (4) linear regulation. A 230/9 V, 1 A transformer is selected so that the secondary voltage is low enough for safe low-voltage circuitry while still providing sufficient headroom for the 5 V regulator. A similar 230/9 V transformer + bridge + 7805 arrangement is documented in an IIT Bombay electronics project.

The LM7805/LM340 family provides fixed 5 V regulation and includes current limiting and thermal shutdown. TI specifies operation up to 35 V input and output current capability up to 1.5 A with appropriate thermal management.

3. Circuit Diagram



CIRCUIT OPERATION

- 230V AC mains is stepped down to 9V AC by T1.
- Bridge rectifier converts AC to pulsating DC.
- C1 filters the pulsating DC and reduces ripple.
- LM7805 regulates the voltage to a steady 5V DC.
- The load (50Ω) receives a constant 5V DC.

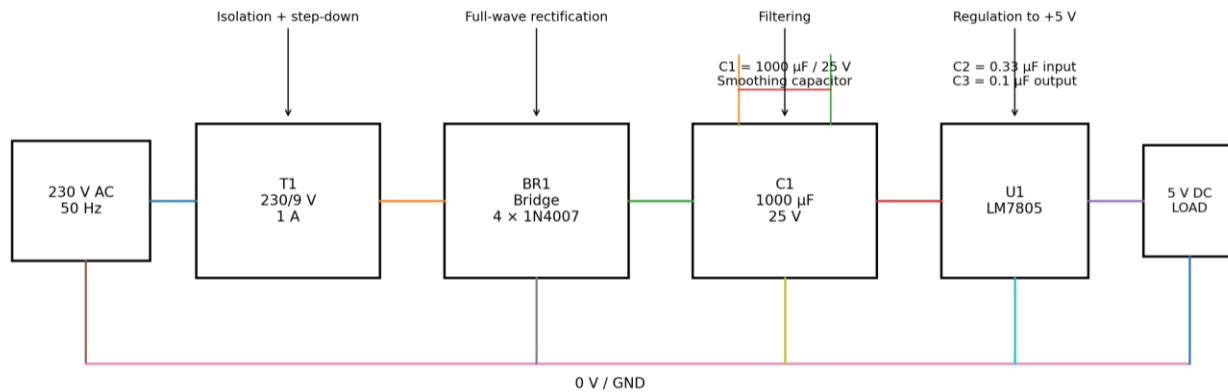


Figure 1. Proposed 230 V AC to 5 V DC regulated power-supply circuit.

4. Components Required

Component	Specification	Qty.	Function	Selection Reason
Step-down transformer	230 V / 9 V AC, 1 A	1	Isolation and voltage reduction	Provides safe low-voltage AC and sufficient current margin
Rectifier diode	1N4007	4	Full-wave bridge rectification	Common, inexpensive rectifier diode with adequate reverse-voltage rating
Filter capacitor	1000 µF / 25 V electrolytic	1	Smooths rectified DC	Reduces 100 Hz ripple at the expected load
Voltage regulator	LM7805 / LM340-5	1	Regulates DC to 5 V	Fixed 5 V output with protection features
Input bypass capacitor	0.33 µF	1	High-frequency bypass	Improves regulator input behavior
Output bypass capacitor	0.1 µF	1	Noise/transient suppression	Improves transient response
Load	50 Ω nominal	1	Represents 100 mA load	Provides a realistic low-power test condition
Fuse	250–500 mA slow-blow, mains-rated	1	Primary-side protection	Recommended safety component for practical construction

Note: The mains-side fuse, switch, enclosure, earthing and proper insulation are essential for a real hardware build. The circuit should not be constructed or tested directly on a breadboard at 230 V.

5. Working Principle

Transformer: The 230 V AC mains is reduced to approximately 9 V RMS AC. The transformer also provides galvanic isolation between the mains and the low-voltage circuitry.

Bridge rectifier: Four 1N4007 diodes form a full-wave bridge. Two diodes conduct in each half cycle, giving a rectified waveform at twice the mains frequency, i.e. 100 Hz.

Filter capacitor: C1 charges near the peak of the rectified waveform and discharges through the load between peaks. This converts pulsating DC into a smoother DC voltage.

Voltage regulator: The LM7805 receives the filtered DC and maintains approximately 5 V at its output as long as its input remains above the required regulation/dropout margin.

Load: The regulated 5 V output is delivered to the load. C2 and C3 provide local bypassing and help reduce transient/noise effects.

6. Design Calculations

- Transformer secondary peak voltage: $V_{pk} = 9 \times \sqrt{2} = 12.73 \text{ V}$.
- Approximate bridge drop: $2 \times 0.7 = 1.4 \text{ V}$.
- Peak voltage after bridge: $V_{pk,rect} \approx 12.73 - 1.4 = 11.33 \text{ V}$.
- Rectifier ripple frequency: $f_r = 2 \times 50 = 100 \text{ Hz}$.
- For a 1000 μF capacitor and 100 mA load, $\Delta V_r \approx I/(f_r C) = 0.1/(100 \times 0.001) \approx 1.0 \text{ V}$ peak-to-peak.
- Approximate minimum capacitor voltage at 100 mA: $11.33 - 1.0 \approx 10.33 \text{ V}$, leaving substantial headroom for a 5 V linear regulator.
- Approximate regulator dissipation at 100 mA: $P_{reg} \approx (10.8 - 5) \times 0.1 \approx 0.58 \text{ W}$. A small heatsink may be considered depending on enclosure and ambient temperature.

7. Simulation Method

A time-domain numerical simulation was performed using a 230 V, 50 Hz source represented by an ideal 230/9 V transformer, a 1.4 V total bridge-diode drop, a 1000 μF smoothing capacitor, a 50 Ω load and a simplified LM7805 regulator model. The regulator model limits the output to 5 V whenever adequate input headroom is available. This model is intended for functional and performance analysis; a transistor-level SPICE model should be used when exact semiconductor losses and temperature are required.

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C:\Users\dell>cd C:\Users\dell\Desktop\python

C:\Users\dell\Desktop\python>python power_supply_simulation.py
Matplotlib is building the font cache; this may take a moment.

===== POWER SUPPLY SIMULATION =====
Transformer secondary RMS : 9.00 V
Secondary peak voltage    : 12.73 V
Bridge peak voltage       : 11.33 V
Filtered DC voltage       : 10.50 V
Regulated output voltage  : 5.00 V
Load resistance          
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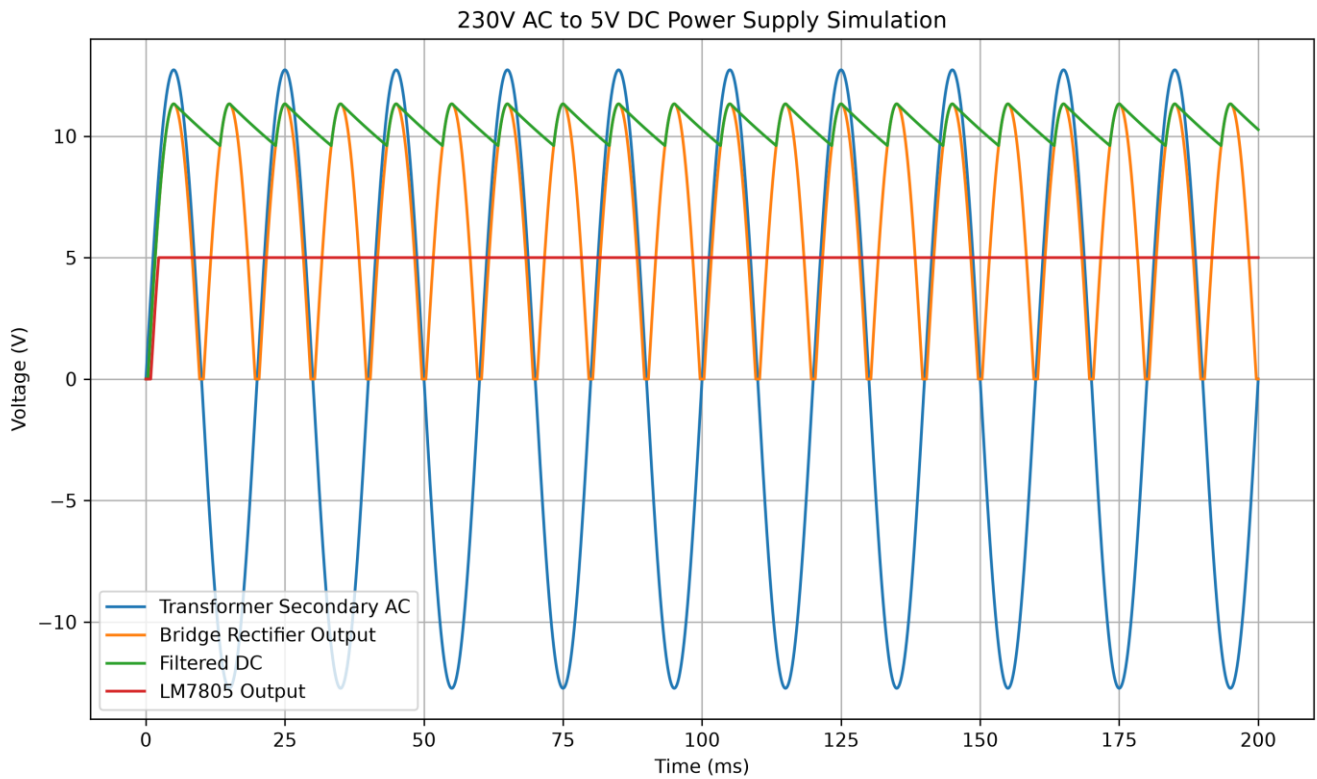


Figure 2. Simulated rectified, filtered and regulated waveforms.

For this project, the **simulation was performed using Python-based numerical time-domain analysis**. The circuit was mathematically modeled stage by stage to study the conversion of 230 V AC into regulated 5 V DC.

Simulation procedure

1. AC input generation:

A 50 Hz sinusoidal waveform representing the 9 V RMS transformer secondary was generated in Python. Its peak voltage is:

$$V_{\text{peak}} = 9 \times \sqrt{2} = 12.73 \text{ V}$$

2. Bridge rectification:

The four 1N4007 diodes were represented mathematically as a full-wave bridge rectifier. Two diode drops of approximately 0.7 V were considered during conduction.

3. Capacitor filtering:

A 1000 μF capacitor and 50 Ω load were modeled to simulate charging and discharging of the capacitor. The resulting waveform represents the filtered DC voltage and its ripple.

4. Voltage regulation:

The LM7805 was represented using a simplified regulator model that limits the output voltage to approximately **5 V DC**.

5. Load analysis:

For a 50 Ω load: $I = 0.1 \text{ A}$. Therefore, the simulated load current is approximately **100 mA**.

6. Waveform analysis:

Python's **NumPy** library was used for numerical calculations, while **Matplotlib** was used to plot the transformer AC waveform, rectified waveform, filtered DC waveform, capacitor ripple, and regulated 5 V output.

Software Used

- **Python 3.14**
- **NumPy**
- **Matplotlib**

The simulation results are then compared with the theoretical calculations to verify the performance of the designed power supply.

8. Simulation Results

Parameter	Simulated / Calculated Result
Transformer secondary	9 V RMS AC
Secondary peak	12.73 V
Bridge output peak	11.33 V
Filtered capacitor maximum	11.33 V
Filtered capacitor minimum	9.61 V
Filter ripple	1.72 V p-p
Regulated output average	5.000 V
Regulated output range	5.000–5.000 V
Test load	50 Ω $\approx$ 100 mA at 5 V

9. Load and Ripple Analysis

Load	Ripple (p-p)	Estimated V _{cap,min}	Regulator dissipation	5 V regulation margin
50 mA	0.50 V	10.83 V	0.30 W	Adequate
100 mA	1.00 V	10.33 V	0.58 W	Adequate
250 mA	2.50 V	8.83 V	1.27 W	Adequate
500 mA	5.00 V	6.33 V	1.91 W	Marginal

The analysis shows that increasing load current increases capacitor ripple approximately in proportion to current. At 500 mA, the idealized ripple estimate reaches about 5 V p-p, which substantially increases regulator heating and reduces practical margin. Therefore, a 9 V transformer + 7805 design is best treated as a low-power supply unless thermal and transformer ratings are carefully verified.

10. Performance Analysis

- **Voltage regulation:** The regulator provides a nominal 5 V DC output with much smaller variation than the unregulated rectifier output.
- **Ripple:** The 1000 μ F capacitor shifts the 100 Hz rectified waveform toward DC. Larger capacitance reduces ripple but increases inrush current and physical size.
- **Efficiency:** A linear regulator dissipates the voltage difference as heat, so efficiency is inherently lower than a modern switching converter.

- Thermal behavior: At higher load current, regulator power dissipation rises quickly. TI documents thermal shutdown and current limiting for the LM7805 family.
- Isolation and safety: The transformer provides electrical isolation from the mains, which is a major safety advantage over transformerless supplies.

11. Advantages and Limitations

Advantages

- Simple circuit and easy to understand
- Galvanic isolation from 230 V mains
- Stable 5 V regulated output
- Low component count
- Suitable for educational and low-power embedded circuits

Limitations

- Linear regulator wastes excess voltage as heat
- Transformer makes the supply relatively bulky
- Output current is limited by transformer, rectifier, capacitor ripple and regulator thermal constraints
- Not intended as a high-efficiency USB/mobile charger without additional regulation and protection

12. Suggested EDA Implementation

The circuit can be entered in KiCad, Proteus, LTspice or another SPICE-capable EDA package. Use the same topology and nominal values given in this report. For a SPICE run, observe the transformer secondary waveform, bridge output, capacitor voltage and regulated 5 V output. A transient analysis of at least 100–200 ms is recommended so that several 50 Hz cycles and the 100 Hz ripple are visible.

13. Conclusion

A practical 230 V AC to 5 V DC linear power supply was designed using a 230/9 V transformer, 1N4007 bridge rectifier, 1000 μ F smoothing capacitor and LM7805 regulator. The calculated and numerically simulated results show that the circuit can provide a regulated output close to 5 V under a representative 100 mA load, with the transformer and capacitor providing sufficient input headroom. The main design trade-off is thermal loss in the linear regulator, especially at higher load current. For applications requiring higher efficiency, a switch-mode buck converter following the isolated transformer stage would be a better alternative.

14. References

1. **Texas Instruments – LM7805 / LM340 Family Datasheet**
Covers the LM7805 5 V regulator, electrical characteristics, protection features, pin configuration and application circuit.
[TI LM7805 Datasheet](#)
2. **Texas Instruments – LM7800 Product Page**
Provides official LM7805 product specifications, including input-voltage range, output current and thermal/short-circuit protection.
[TI LM7800 Product Page](#)

3. **Vishay – 1N4001–1N4007 Rectifier Diodes Datasheet**

Useful for the bridge-rectifier component specifications and ratings.

[Vishay 1N4007 Datasheet](#)

4. **Vishay – 1N4001–1N4007 Product Information**

Provides applications and general characteristics of the 1N4007 family for power-supply rectification.

[Vishay 1N4007 Product Page](#)